

Section 1 - Practice Problems

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Chapter 1.1: Sets and Functions

Exercise 1.1.5 Prove the following distributive laws:

(a) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.

Solution:

We can prove that two sets are equal by showing that they are both subsets of one another.

(I) Show that $A \cap (B \cup C) \subseteq (A \cap B) \cup (A \cap C)$.

Let $x \in (A \cap (B \cup C))$. Therefore, by definition of intersection and union, we know $x \in A \wedge x \in (B \cup C)$. Thus, this implies that $x \in B \vee x \in C$. However, $x \in A \implies x \in ((A \cap B) \cup (A \cap C)) \iff x \in (B \cup C)$. However, we already determined that $x \in B \vee x \in C$. Thus, $x \in (A \cap (B \cup C)) \implies x \in ((A \cap B) \cup (A \cap C))$ and therefore $A \cap (B \cup C) \subseteq (A \cap B) \cup (A \cap C)$.

(II) Show that $A \cap (B \cup C) \supseteq (A \cap B) \cup (A \cap C)$.

Let $x \in ((A \cap B) \cup (A \cap C))$. Therefore, by definition of intersection and union, we know $(x \in A \wedge x \in B) \vee (x \in A \wedge x \in C)$. This clearly implies that $x \in A$, and furthermore that $x \in B \vee x \in C$. Therefore, we know $x \in (B \cup C)$, and since $x \in A$, $x \in A \cap (B \cup C)$. Thus, $x \in ((A \cap B) \cup (A \cap C)) \implies x \in (A \cap (B \cup C))$ and thus $A \cap (B \cup C) \supseteq (A \cap B) \cup (A \cap C)$.

Therefore, since $A \cap (B \cup C) \subseteq (A \cap B) \cup (A \cap C)$ and $A \cap (B \cup C) \supseteq (A \cap B) \cup (A \cap C)$, $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.

(b) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.

Solution:

We will prove this in much the same way as in (a).

(III) Show that $A \cup (B \cap C) \subseteq (A \cup B) \cap (A \cup C)$.

Let $x \in (A \cup (B \cap C))$. Therefore, $x \in A \vee x \in (B \cap C)$. If $x \in (B \cap C)$ which means we know that $x \in B \wedge x \in C$. Therefore, we know that $x \in A \vee (x \in B \wedge x \in C)$. If $x \in A$, then $x \in (A \cup B) \wedge x \in (A \cup C)$, and therefore $x \in ((A \cup B) \cap (A \cup C))$, and if $x \in B \wedge x \in C$, then $x \in (A \cup B) \wedge x \in (A \cup C)$ and therefore $x \in ((A \cup B) \cap (A \cup C))$. Therefore, $x \in (A \cup (B \cap C)) \implies x \in ((A \cup B) \cap (A \cup C))$ and thus $A \cup (B \cap C) \subseteq ((A \cup B) \cap (A \cup C))$.

(IV) Show that $A \cup (B \cap C) \supseteq (A \cup B) \cap (A \cup C)$.

Let $x \in ((A \cup B) \cap (A \cup C))$. Therefore, $x \in (A \cup B) \wedge x \in (A \cup C)$ which means we know that $x \in A \vee (x \in B \wedge x \in C)$. If $x \in A$, then trivially $x \in (A \cup (B \cap C))$, and if $x \in B \wedge x \in C$, then $x \in (B \cap C)$ and therefore $x \in (A \cup (B \cap C))$. Therefore $x \in ((A \cup B) \cap (A \cup C)) \implies x \in (A \cup (B \cap C))$ and thus $(A \cup B) \cap (A \cup C) \subseteq A \cup (B \cap C)$.

Therefore, since $A \cup (B \cap C) \subseteq (A \cup B) \cap (A \cup C)$ and $A \cup (B \cap C) \supseteq (A \cup B) \cap (A \cup C)$, $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.

Exercise 1.1.11 Let $g(x) := x^2$ and $f(x) := x + 2$ for $x \in \mathbb{R}$, and let h be the composite function $h := g \circ f$.

(a) Find the direct image $h(E)$ of $E := \{x \in \mathbb{R} : 0 \leq x \leq 1\}$.

Solution:

Recall that the image of a function under a set is the set of all values which the function can attain with inputs taken from the set. We know the composite function h is given by $h(x) = f(x)^2 = (x + 2)^2$. Therefore, the image of h under E is $\{y \in \mathbb{R} : 4 \leq y \leq 9\}$.

(b) Find the inverse image $h^{-1}(G)$ of $G := \{x \in \mathbb{R} : 0 \leq x \leq 4\}$.

Solution:

Recall that the inverse image of a function is the set of all values in the domain of the function which are in the given set. Since we previously determined that $h(x) = (x + 2)^2$, by rearranging, we can find that:

$$\begin{aligned} x &= (h^{-1}(x) + 2)^2 \\ \pm\sqrt{x} &= (h^{-1}(x) + 2) \\ -2 \pm \sqrt{x} &= h^{-1}(x) \end{aligned}$$

Therefore, $h^{-1}(x) = -2 \pm \sqrt{x}$. Thus, the inverse image of h under G is $\{x \in \mathbb{R} : -2 \leq x \leq 2\}$.

Exercise 1.1.19

(a) Show that if $f : A \rightarrow B$ is injective and $E \subseteq A$, then $f^{-1}(f(E)) = E$. Give an example to show that equality need not hold if f is not injective.

Solution:

Proof. Let $x \in E$. By definition of inverse, we know trivially that $x \in E \implies x \in f^{-1}(f(E))$ and thus $E \subseteq f^{-1}(f(E))$. Now consider $x \in f^{-1}(f(E))$. To prove that x must necessarily be in E , we rely on a proof by contradiction. Assume for the purposes of contradiction that $x \notin f^{-1}(f(E))$. Since we are taking the inverse of $f(E)$, that implies $\exists y \in E$ such that $f(y) = f(x)$. But since $x \notin E$ and $y \in E$, we necessarily know that $x \neq y$. However, this would violate the assumption that f is injective since, if this were true $\exists x, y \in A, x \neq y$ such that $f(x) = f(y)$. Thus, by contradiction, $x \in f^{-1}(f(E)) \implies x \in E$ and $f^{-1}(f(E)) \subseteq E$.

Therefore, since $E \subseteq f^{-1}(f(E))$ and $f^{-1}(f(E)) \subseteq E$, $E = f^{-1}(f(E))$ for injective f . $\square$

If injectivity is not assumed, this is not necessarily true. Taking $f(x) = x^2$ and $E = [0, 1]$ as an example, $f^{-1}(f(E)) = [-1, 1]$. Clearly then, for non-injective f , equality need not hold.

- (b) Show that if $f : A \rightarrow B$ is surjective and $H \subseteq B$, then $f(f^{-1}(H)) = H$. Give an example to show that equality need not hold if f is not surjective.

Solution:

Proof. Let $K = f^{-1}(H)$. Since f is surjective, we know that $f(K) = H$ and since $K = f^{-1}(H)$, then $H = f(f^{-1}(H))$. Thus, if $x \in f(f^{-1}(H))$, then $x \in H$ and therefore $H \subseteq f(f^{-1}(H))$. $\square$

Kind of a gross counter example but i'm tired sorry.

If surjectivity is not assumed, this is not necessarily true. Once again, take $f(x) = x^2$ and $H = (-\infty, \infty)$. $f(f^{-1}(H)) = [0, \infty)$. Clearly then, for non-surjective f , equality need not hold.

Exercise 1.1.21 Prove that if $f : A \rightarrow B$ and $g : B \rightarrow C$ are bijective, then the composite $g \circ f$ is a bijective map of A onto C .

Solution:

Proof. To prove that $g \circ f$ is a bijective map of A onto C , we must show that $g \circ f$ is both injective and surjective.

(I) Prove $g \circ f$ is injective.

Since it is given that f and g are injective, then we know $\forall i, j \in A, i \neq j, f(i) \neq f(j)$ and $\forall k, l \in B, g(k) \neq g(l)$. Thus:

$$\begin{aligned}\forall x, y \in A, x \neq y &\implies \\ f(x) \neq f(y) &\implies \\ g(f(x)) \neq g(f(y))\end{aligned}$$

Therefore, $g \circ f$ is an injective map of A onto C .

(II) Prove $g \circ f$ is surjective.

Since it is given that f and g are surjective, $i \in B \implies \exists j \in A$ such that $f(j) = i$ and $k \in C \implies \exists l \in B$ such that $g(l) = k$. Thus:

$$\begin{aligned}x \in C &\implies \\ \exists y \in B \text{ such that } g(y) = x &\implies \\ \exists z \in A \text{ such that } f(z) = y &\implies \\ g(f(z)) = x\end{aligned}$$

Therefore $g \circ f$ is a surjective map of A onto C .

Thus, since $g \circ f$ is both an injective and surjective map of A onto C , $g \circ f$ is bijective.

□

Chapter 1.2: Mathematical Induction

Exercise 1.2.1 Prove that $\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \dots + \frac{1}{n \cdot (n+1)} = \frac{n}{n+1} \quad \forall n \in \mathbb{N}$.

Solution:

Proof. Let $P(n)$ be the statement $P(n) : \frac{1}{1} \cdot 2 + \frac{1}{2} \cdot 3 + \dots + \frac{1}{n \cdot (n+1)} = \frac{n}{n+1}$.

(I) Show the base case holds (Prove $P(1)$ is true).

LHS:	RHS:
$\frac{1}{n \cdot (n+1)}$	$\frac{n}{n+1}$
$= \frac{1}{1 \cdot (1+1)}$	$= \frac{1}{1+1}$
$= \frac{1}{2}$	$= \frac{1}{2}$

Thus, $P(1)$ is true.

(II) State induction hypothesis (State $P(k)$).

Assuming there exists a $k \in \mathbb{N}$ for which $P(k)$ is true we can formulate our induction hypothesis as:

$$P(k) : \frac{1}{1} \cdot 2 + \frac{1}{2} \cdot 3 + \dots + \frac{1}{k \cdot (k+1)} = \frac{k}{k+1}$$

(III) Prove $P(k) \implies P(k+1)$.

$$\begin{aligned}
 & \frac{1}{1} \cdot 2 + \frac{1}{2} \cdot 3 + \dots + \frac{1}{k \cdot (k+1)} + \frac{1}{(k+1) \cdot (k+2)} \\
 &= \frac{k}{k+1} + \frac{1}{(k+1) \cdot (k+2)} \\
 &= \frac{k(k+2) + 1}{(k+1) \cdot (k+2)} \\
 &= \frac{k^2 + 2k + 1}{(k+1) \cdot (k+2)} \\
 &= \frac{(k+1)^2}{(k+1) \cdot (k+2)} \\
 &= \frac{k+1}{k+2}
 \end{aligned}$$

Therefore, since $P(1)$ is true, and $P(k) \implies P(k+1)$ for arbitrary $k \in \mathbb{N}$, $P(n)$ is true $\forall n \in \mathbb{N}$. $\square$

Exercise 1.2.7 Prove that $5^{2n} - 1$ is divisible by 8 $\forall n \in \mathbb{N}$.

Solution:

Proof. Let $P(n)$ be the statement $P(n) : 8 \mid (5^{2n} - 1)$.

(I) Show the base case holds (Prove $P(1)$ true).

$$\begin{aligned}
 8 \mid (5^{2n} - 1) &= \text{mod}(5^{2 \cdot 1} - 1) \\
 &= 8 \mid (5^2 - 1) \\
 &= 8 \mid (24) \\
 &= 0
 \end{aligned}$$

Therefore, $P(1)$ is true.

(II) State induction hypothesis (State $P(k)$).

Assuming there exists a $k \in \mathbb{N}$ for which $P(k)$ is true we can formulate our induction hypothesis as:

$$P(k) : 8 \bmod (5^{2k} - 1)$$

(III) Prove $P(k) \implies P(k+1)$.

$$\begin{aligned} 8 \bmod (5^{2(k+1)} - 1) &= 8 \bmod (5^{2k+2} - 1) \\ &= 8 \bmod (5^2 \cdot 5^{2k} - 1) \\ &= 0 \end{aligned}$$

Therefore, since $P(1)$ is true, and $P(k) \implies P(k+1)$ for arbitrary $k \in \mathbb{N}$, $P(n)$ is true $\forall n \in \mathbb{N}$. $\square$

Exercise 1.2.11 Conjecture a formula for the sum of the first n odd natural numbers $1 + 3 + \dots + (2n - 1)$, and prove your formula using mathematical induction

Solution:

First n odd natural numbers	1	2	3	4	5	...
$1 + 3 + \dots + (2n - 1)$	1	4	9	16	25	...

Based on this, I conjecture that $1 + 3 + \dots + (2n - 1) = n^2$. Next, I will prove this fact.

Proof. Let $P(n)$ be the statement $P(n) : 1 + 3 + \dots + (2n - 1) = n^2$.

(I) Show base case holds (Prove $P(1)$ true).

LHS:	RHS:
$(2n - 1)$	n^2
$= (2(1) - 1)$	$(1)^2$
$= 1$	$= 1$

Therefore, $P(1)$ is true.

(II) State induction hypothesis (State $P(k)$).

Assuming there exists a $k \in \mathbb{N}$ for which $P(k)$ is true we can formulate our induction hypothesis as:

$$P(k) : 1 + 3 + \dots + (2k - 1) = k^2$$

(III) Prove $P(k) \implies P(k+1)$

$$\begin{aligned} 1 + 3 + \dots + (2k-1) + (2k+1) &= k^2 + (2k+1) \\ &= (k+1)^2 \end{aligned}$$

Therefore, since $P(1)$ is true, and $P(k) \implies P(k+1)$ for arbitrary $k \in \mathbb{N}$, $P(n)$ is true $\forall n \in \mathbb{N}$. $\square$

Exercise 1.2.13 Prove that $n < 2^n \forall n \in \mathbb{N}$.

Solution:

Let $P(n)$ be the statement $P(n) : n < 2^n$.

Proof. (I) Show base case holds (Prove $P(1)$ true).

LHS:	RHS:
n	2^n
$= (1)$	$= 2^1$
$= 1$	$= 2$

Therefore, $P(1)$ is true.

(II) Formulate induction hypothesis (State $P(k)$).

Assuming there exists a $k \in \mathbb{N}$ for which $P(k)$ is true we can formulate our induction hypothesis as:

$$P(k) : k < 2^k$$

(III) Prove $P(k) \implies P(k+1)$

$$\begin{aligned} k+1 &< 2^k + 1 \\ &< 2^{k+1} \end{aligned}$$

Therefore, since $P(1)$ is true, and $P(k) \implies P(k+1)$ for arbitrary $k \in \mathbb{N}$, $P(n)$ is true $\forall n \in \mathbb{N}$. $\square$

Chapter 1.3: Finite and Infinite Sets

Exercise 1.3.3 Let $S := \{1, 2\}$ and $T := \{a, b, c\}$

(a) Determine the number of different injections from S into T .

Solution:

My answer is 6 and my reasoning is as follows:

Taking $s_1 \in S$, we have three possible choices of $t_1 \in T$ for which we can create an injection from s_1 to T , giving us 3 different injections. Now taking $s_2 \in S$, we would have three possible choices of $t_2 \in T$, but since the relation must be injective, t_1 cannot equal t_2 and thus we only have two possible choices. Thus, we have $3 \cdot 2 = 6$ possible injections from S to T .

(b) Determine the number of different surjections from T onto S .

Solution:

My answer is 6 and my reasoning is as follows:

Taking $t_1 \in T$, t_1 has two possible choices of $s_1 \in S$ for which we can create a surjection from t_1 to S . Now consider $t_2 \in T$. Since the relationship need not be injective, we separate our choices into two possible cases. If the surjection from t_2 to S is s_1 , then we know that t_3 MUST be s_2 . However, if the surjection from t_2 to S is not s_1 , then t_3 can relate to either of s_1 or s_2 . Thus, we have $2 \cdot 1 \cdot 1 + 2 \cdot 1 \cdot 2 = 6$ possible surjections from T to S .

Exercise 1.3.5 Give an explicit definition of the bijection f from $\mathbb{N}$ onto $\mathbb{Z}$ described in example 1.3.7.(b).

Solution:

$$f(n) = \begin{cases} 0, & \text{if } n = 1 \\ \frac{n}{2}, & \text{if } n \text{ is even} \\ -\frac{n-1}{2}, & \text{if } n \text{ is odd and } n \neq 1 \end{cases}$$

Exercise 1.3.7 Prove that a set T_1 is denumerable if and only if there is a bijection from T_1 onto a denumerable set T_2 .

Solution:

Proof. We prove this by first independently proving the forward and reverse directions of this statement.

- (I) Prove the forward direction (T_1 is denumerable $\implies$ there exists a bijection from T_1 to a denumerable set T_2).

Since both T_1 and T_2 are denumerable, we know that there exists two bijection functions f, g such that $f : \mathbb{N} \rightarrow T_1$ and $g : \mathbb{N} \rightarrow T_2$ respectively. However, we know that g^{-1} is also a bijective function from T_2 onto $\mathbb{N}$, and from exercise 1.1.21, we know that the composite function $g^{-1} \circ f$ is a bijection from T_1 onto T_2 . Thus, if T_1 and T_2 are both denumerable, there must exist a bijection from T_1 to T_2 .

- (II) Prove the reverse direction (A bijection from T_1 to a denumerable set $T_2 \implies T_1$ is denumerable),

We know that there exists a bijection from T_1 to T_2 . Let's call such a bijection, $f : T_1 \rightarrow T_2$. Since T_2 is denumerable, we know that there exists a bijection from T_2 onto $\mathbb{N}$. Let's call that bijection $g : T_2 \rightarrow \mathbb{N}$. However, from exercise 1.1.21, we proved that $g \circ f$ would be a bijective map from T_1 onto $\mathbb{N}$, making $(f \circ g)^{-1}$ a bijection from $\mathbb{N}$ onto T_1 . Thus, by definition of denumerability, T_1 is denumerable.

Thus, since both the forward and reverse directions are true, T_1 is denumerable $\iff$ there exists a bijection from T_1 to a denumerable set T_2 . $\square$

Exercise 1.3.13 Prove that the collection $\mathcal{F}(\mathbb{N})$ of all *finite* subsets of $\mathbb{N}$ is countable.

Solution:

Proof. Note: this proof is technically incorrect since certain numbers, such as 4, map to the singleton set $\{2\}$, which is undesirable, but I feel like this approach is in the right direction. Will update with a more correct proof in future.

To prove that $\mathcal{F}$ is bijective, consider the fact that every natural number (besides 1) can be decomposed into the product of primes. Using this fact, we can construct a bijection between $\mathbb{N}$ and $\mathcal{F}$ with the function:

$$f(n) = \begin{cases} \{1\} & \text{if } n = 1 \\ \{i + 1\} & \text{if } n \text{ is the } i^{\text{th}} \text{ prime number.} \\ \{j, k\} & \text{if } n \text{ has two prime factors } n_1, n_2 \text{ and where} \\ & n_1, n_2 \text{ are the } j^{\text{th}} \text{ and } k^{\text{th}} \text{ prime numbers respectively.} \\ \vdots & \end{cases}$$

Thus, since there exists a bijective mapping from $\mathbb{N}$ to $\mathcal{F}$, by definition of denumerability, $\mathcal{F}$ is denumerable and is thus countable. $\square$